Session 1

Q: 01 Name 5 IT job roles and write one line about what each role does.

1) Developer -> Writes and maintains software applications.
2) Designer -> Designs attractive and user-friendly interfaces.
3) Testing ->Tests software to find bugs and ensure quality.
4) Data Analyst ->Analyzes data to help businesses make decisions.
5) AI engineer -> Builds and trains Artificial Intelligence and Machine Learning models.

Q: 02 List 3 product companies and 3 service companies with one example of work each does.

Product Companies:-
Google:
Example of work: Develops Android, Search, Gmail, and Google Maps.

Microsoft:
Example of work: Develops Windows, Microsoft Office, and Azure Cloud.

Zoho: Creates business software like Zoho CRM and Zoho Mail.

Service Companies:-

Tata Consultancy Services (TCS):-
Example of work: Develops Android, Search, Gmail, and Google Maps.

Wipro:-
Example of work: Provides cloud, AI, cyber security, and software services.

Infosys:-
Example of work: Develops software and digital transformation solutions.

Q: 03 Write the full forms and one-line use of GitHub, Slack, Jira, IDE, API.

GitHub:-
Full Form:- Git+Hub(Git is a distributed version control system, and "Hub" is a center of activity)
Use:- Stores source code and helps developers collaborate.

Slack:-
Full Form:- Searchable Log of All Conversation and Knowledge
Use:- Team communication and messaging platform.

Jira:-
Full Form:- Jira (official product name, not an acronym)
Use:- Used to track tasks, bugs, and software projects.

IDE:-
Full Form:- Integrated Development Environment
Use:- Used to write, run, and debug code.

API:-
Full Form:- Application Programming Interface
Use:- Allows different software applications to communicate.

Q: 04 Name any 5 programming languages and mention one popular use case for each.

Programming Language:-

C
Use :- System programming and embedded systems

Java
Use:- Android apps and enterprise software

Python
Use:- AI, Machine Learning, and Data Science

Kotlin
Use:- Android app development

JavaScript
Use:- Interactive websites and web applications

Q: 05 What is the difference between a developer and a tester? (2 lines)

Developer: Develops software by writing and maintaining code.

Tester: Tests the software to find bugs and ensure it works correctly before release.

Session 2

Software Applicatio:-

A **software application** is a program that helps users perform specific tasks, such as browsing the internet, editing documents, or chatting.

Q: 01 What are the three main layers of software architecture? Write one line about each.

Below main 3 layers

1) Presentation(UI) layer -> user interact with the software (Frontend)
2) Business Logic -> Processes user requests, applies business rules, and performs calculations.
3) Database -> Stores, retrieves, updates, and manages application data.

Q: 02 Define client and server in your own words with one real-life example.

Client:-

A client is the device or application used by a user to request information or services

Server:-

A server is a computer that receives requests from clients, processes them, and sends back the required data.

Real-Life Example (Web Browser):-

When you open a website (for example, **www.google.com**) in your web browser:

Client: Your web browser (such as Google Chrome, Microsoft Edge, or Firefox).

Server: The web server that hosts the Google website.

How it works: The browser sends a request to the server. The server processes the request, retrieves the required data, and sends the webpage back to the browser, which displays it on your screen.

Q: 03 Write 2 differences each between desktop apps, web apps, and mobile apps.

Desktop App:-

Installation:- Must be installed on a computer.

Device:- Runs on Windows, macOS, or Linux.

Web App:-

Installation:- No installation needed; runs in a web browser.

Device:- Works on any device with a browser and internet.

Mobile App:-

Installation:- No Installed from an app store.

Device:- Runs on Android or iOS smartphones/tablets.

Q: 04 Name any 3 databases and mention whether each is SQL or NoSQL.

Database	Type
MySQL	SQL
PostgreSQL	SQL
MongoDB	NoSQL

Session 03

Q: 01 Full Form

1) HTTP -> Hyper Text Transfer Protocol
2) HTTPS -> Hyper Text Transfer Protocol Secure
3) FTP -> File Transfer Protocol
4) SMTP -> Simple Mail Transfer Protocol

Q: 02 What is the difference between IPv4 and IPv6? Answer in 2 lines.

IPv4: Uses **32-bit** addresses (e.g., **192.168.1.1**) and supports about **4.3 billion** unique IP addresses.

IPv6: Uses **128-bit** addresses (e.g., **2001:db8::1**) and supports a vastly larger number of unique IP addresses, providing much more address space than IPv4.

Q: 03 Run nslookup google.com on your system and write down the IP address you get.

You need to run the command on **your own computer**, because the IP address returned can vary depending on your location and Google's DNS(Domain Name System) load balancing.

Steps (Windows)

1) Press **Win + R**.

2) Type **cmd** and press **Enter**.

3) Type the following command and press **Enter**:

</> Bash

Command:- nslookup google.com

- **Server:** dns.google
- **IPv4 Address:** 142.251.153.119
- **IPv6 Address:** 2001:4860:482a:7700::

Q:04 HTTP code Status:

1) 200 -> Request Successful
2) 301 -> URL (HTTP) Error
3) 404 -> Not Found
4) 500 -> Server Error
5) 403 -> Forbidden Error

Q: 05 Explain DNS hierarchy (root, TLD, domain, sub-domain) with www.google.com as an example.

. -> Root

www -> sub- domain

google ->Domain

.com -> TLD (Top Level Domain)

Session 04

Q : 01 List all 6 phases of SDLC in order with one line about each.

Requirement Gathering

|
V
Analysis
|
V
Designing
|
V
Implementation
|
V
Testing
|
V
Maintenance

Q:02 Difference between Agile and Waterfall model

Waterfall Model:

- Waterfall model is linear sequential model
- Changes are difficult and costly once a phase is completed
- Planning fixed and predefined
- Slow delivery due to sequential execution

Agile Model:

- Agile model is iterative and increment
- Changes can be easily incorporated at any stage of development
- Planning is flexible and adaptive
- Fast deliver of working features

Q:03 Name the 3 main Scrum roles and write one line about the responsibility of each.

1) Product owner : Maximizes the value of the product by managing are prioritizing the product backlog.
2) Scrum Master : Ensures the scrum team adheres to scrum theory, practices and rules while removing
3) Development Team : Designs, develops, tests and delivers the software.

Q: 04 What is a Sprint in Scrum? Mention its typical duration.

Sprint:-
A **Sprint** is a fixed period during which the Scrum team completes a set of planned tasks or features.

Typical Duration:-
1 to 4 weeks (most commonly **2 weeks**).

Q: 05 What do these Jira board columns mean: To Do, In Progress, Done, Backlog?

Backlog:- List of tasks or user stories that are planned but not started yet.

To Do:- Tasks that are ready to be started.

In Progress:- Tasks that are currently being worked on.

Done:- Tasks that have been completed successfully.

Session 05

Q:-01 Write the Git command for each: initialize a repo, stage all files, commit with a message, push to remote, pull latest changes.

- **Initialize a repository**: git init
- **Stage all files**: git add .
- **Commit with a message**: git commit -m "your message here"
- **Push to remote**: git push *(or git push origin <branch_name> for the initial push)*
- **Pull latest changes**: git pull

Q:-02 What is the difference between git fetch and git pull? Answer in 2 lines.

git fetch: Downloads the latest changes from the remote repository but does **not** merge them into your local branch.

git pull: Downloads the latest changes **and automatically merges** them into your current local branch.

Q:-03 Create a new GitHub repository, push any small text file, and share the repo link.

1) The Quickest Way (Via the Web Browser)
2) Using the Command Line (Git CLI)

1) The Quickest Way (Via the Web Browser):-

You do not need to install anything on your computer to do this.

1) Log in to GitHub.
2) In the upper-right corner of the page, click the **+** drop-down menu and select **New repository**.
3) Give your repository a name (e.g., my-first-repo) and ensure the visibility is set to **Public** so you can share the link.
4) Check the box that says **Add a README file** (this serves as your small text file).
5) Click **Create repository**.
6) Copy the URL from your browser's address bar to share your new repository link.

Q:- 04 What is the difference between a Fork and a Pull Request? Answer in 2 lines.

Fork: Creates your own copy of someone else's GitHub repository in your account.

Pull Request (PR): A request to merge your changes from your branch or fork into the original repository.

Q:-05 Create a new branch named feature-test, make one commit, and merge it back to the main branch.

Create new branch:- git checkout -b feature-test

Commit:- git commit -m "Initial commit"

Merge it back to the main branch:- git merge feature-test